Import Data using Transform Maps (Spreadsheet)

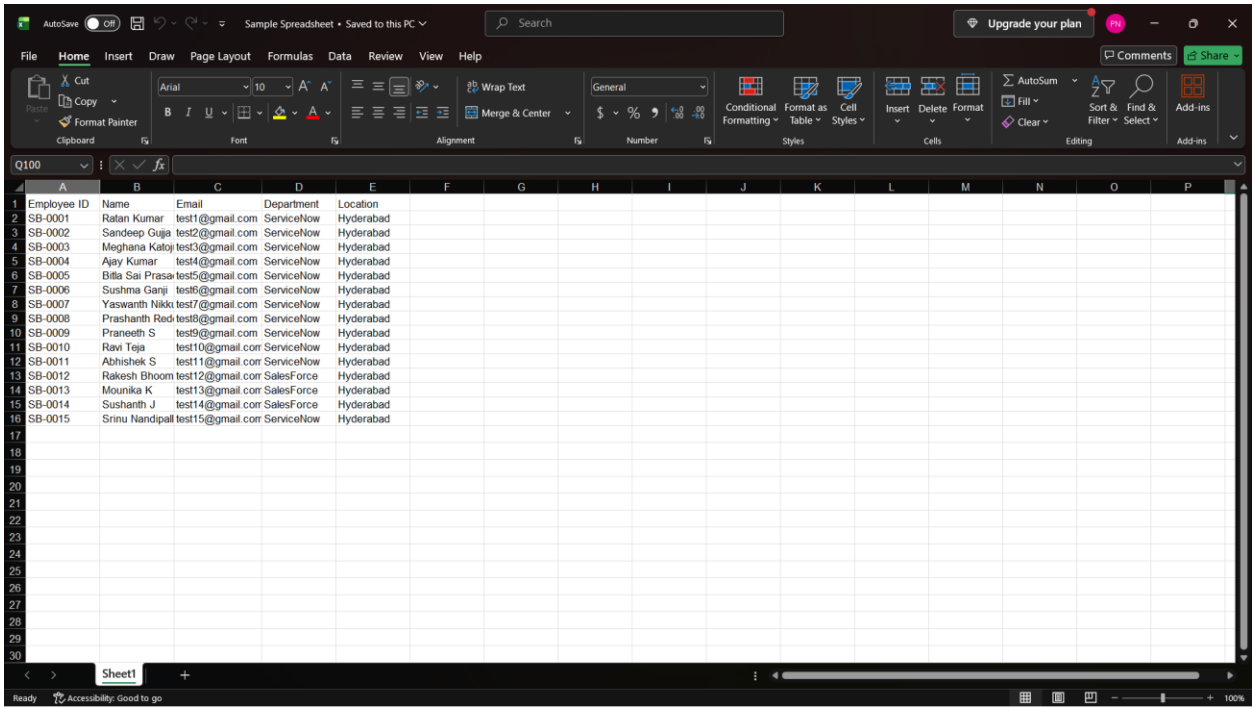
Description:

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is first loaded into an Import Set table, which serves as a staging area, and a Transform Map is then configured to map source fields to the appropriate target table fields. Upon executing the transformation, records are created in the target table and validated to ensure data integrity. This project helps beginners understand the end-to-end data import process in ServiceNow, including data loading, field mapping, transformation execution, result verification, duplicate prevention using Coalesce, and reporting via Dashboards.

Step by Step process:

Phase 1: Prepare the sheets

- **1.** Open New Google Spreadsheet / Microsoft Excel.
- **2.** Create New Spreadsheet with sample employee data (Employee ID, Name, Email, Department, Location).
- **3.** Download the spreadsheet in Sample Spreadsheet.xlsx format.



Employee ID	Name	Email	Department	Location
SB-0001	Ratan Kumar	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	Sandeep Gujja	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	Meghana Katoji	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	Ajay Kumar	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	Bitla Sai Prasa	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	Sushma Ganji	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	Yaswanth Nikki	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	Prashanth Red	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	Praneeth S	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	Ravi Teja	test10@gmail.com	ServiceNow	Hyderabad
SB-0011	Abhishek S	test11@gmail.com	ServiceNow	Hyderabad
SB-0012	Rakesh Bhoom	test12@gmail.com	SalesForce	Hyderabad
SB-0013	Mounika K	test13@gmail.com	SalesForce	Hyderabad
SB-0014	Sushanth J	test14@gmail.com	SalesForce	Hyderabad
SB-0015	Srinu Nandipall	test15@gmail.com	ServiceNow	Hyderabad

Phase 2: Create new Custom table

Navigation:

- Tables > Create New
- Label: Employee Test
- Name: u_employee_test
- After creating and saving the Employee Test table, scroll down to the bottom of the table form.
- Click Show Form to open the table form.
- On the form, locate the Form Context Menu / Additional Actions option on the left side of the form header.
- Click the Form Context Menu / Additional Actions icon.
- Click Configure, select Form Layout.

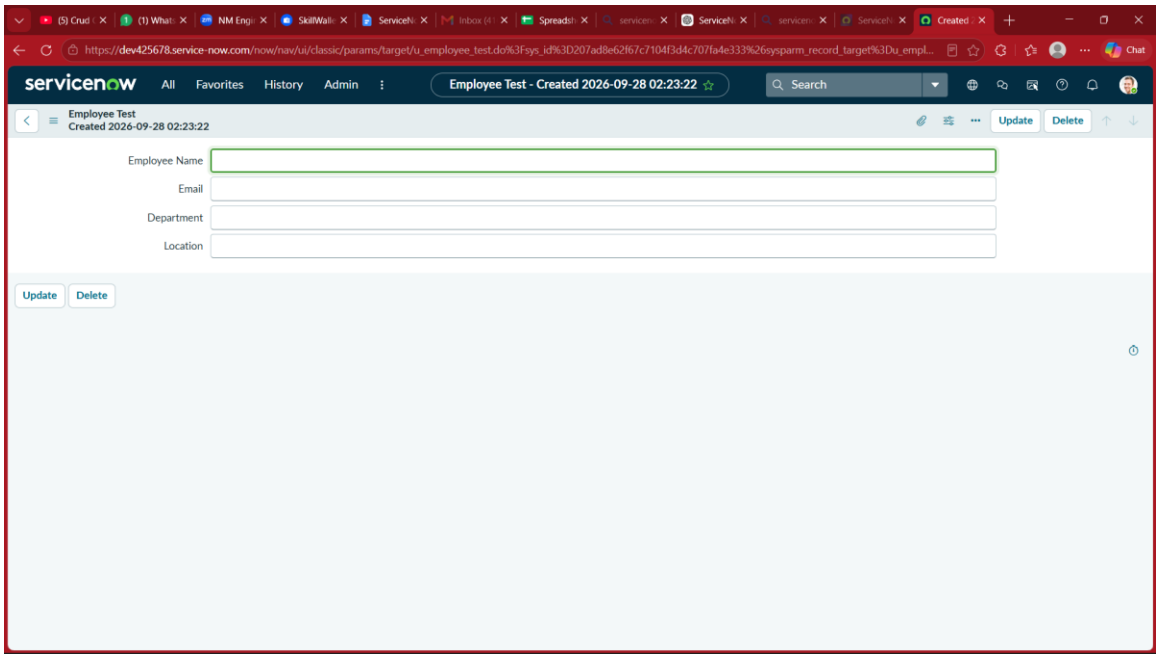


Figure: Employee Test - Table Record Created View

- The Form Layout page will open, where you can configure the fields displayed on the form.
- After opening Form Layout, click New to create a new field.
- Enter the field details:
  - Column Label: Employee ID
  - Type: String
  - Click Add to create the field.
- Repeat the same steps to create the following fields:
  - Employee Name – Type: String
  - Email – Type: String
  - Department – Type: String
  - Location – Type: String

- After creating all the required fields, click Save.
- Verify that the newly created fields are displayed on the Employee Test form.

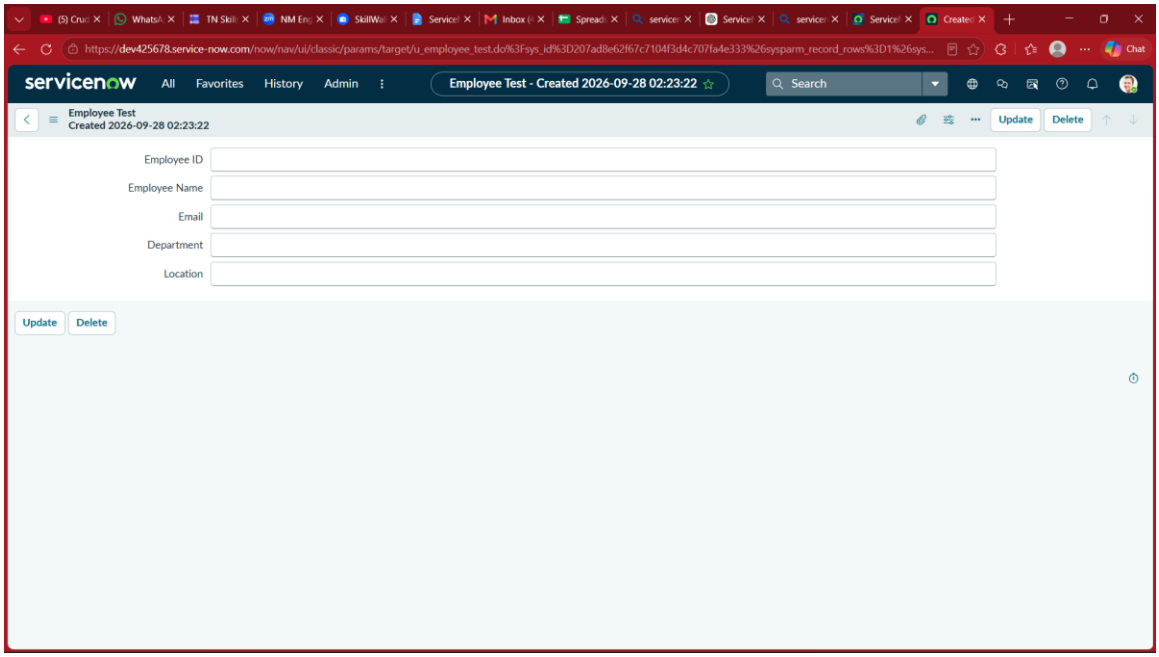


Figure: Employee Test Form with Employee ID and Form Fields Configured

Click on Submit or Save.

Phase 3: Import Set Table

Navigation:

- In Application Navigator → Load Data

Click on Load Data and create an import table. Fill details as shown below:

- Label: Employee Import
- Name: u_employee_import (auto populated)
- Source of the import: File (Sample Spreadsheet.xlsx)
- Click on Submit.

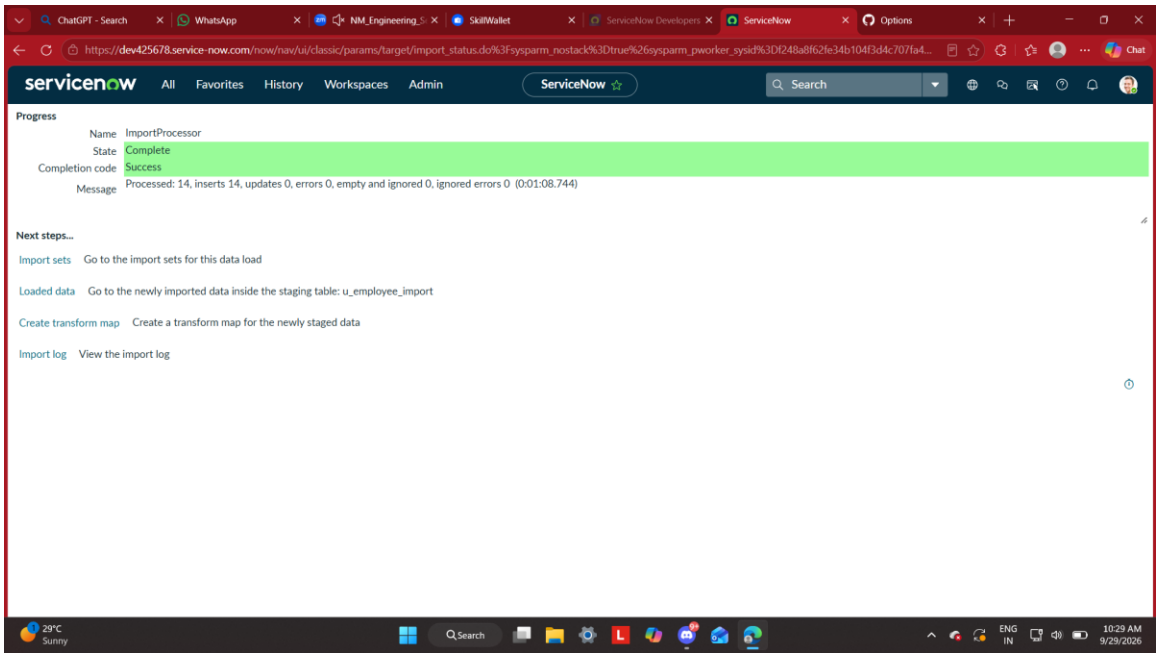


Figure: Import Set Processing Complete - 14 Records Loaded into Staging Table

Once we click on Submit, we can see the success message as shown above (Processed: 14, inserts 14). Also we need to make sure to click on Create Transform Map.

Phase 4: Create Transform Map

In this phase we need to create the Transform Map by clicking on the Create Transform Map button as shown in above screenshot.

Give inputs as shown below:

- Name: Sample Spreadsheet Import
- Source Table: Employee Import [u_employee_import] (auto-populated)
- Target Table: Employee Test [u_employee_test] (as we created earlier)
- Click on Auto Map Matching fields or Mapping Assist to map the fields with Source & Target Table.
- Click on Save.

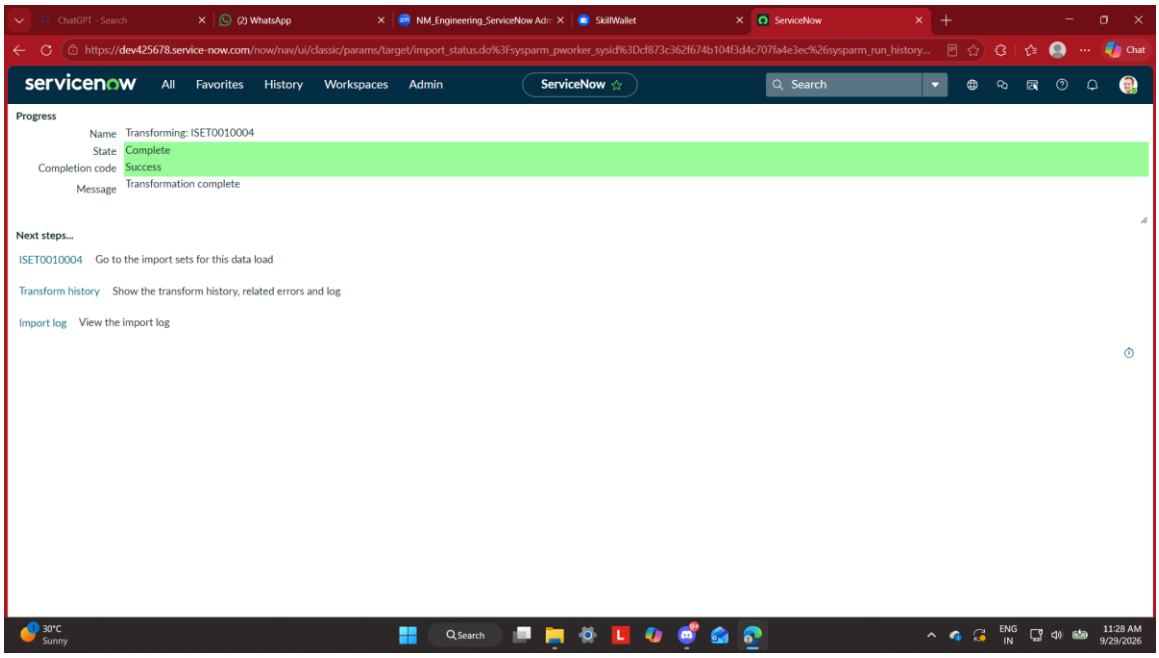


Figure: Transformation Execution Progress - Complete & Success

Make sure both Source & Target table fields are mapped correctly and transformation status shows Complete & Success.

Phase 5: Transform Data & Validate

Navigation:

- Open Employee Test Table from application navigator.
- As our import succeeded, all records are created in the Employee Test Table.

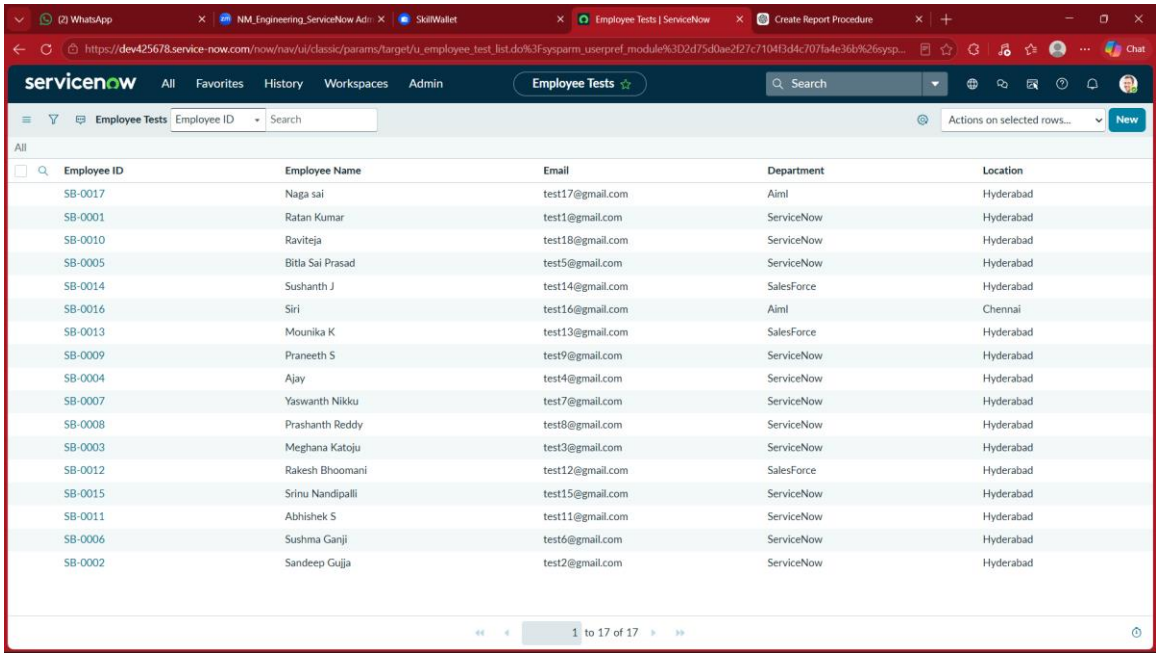


Figure: Employee Tests Target Table View - Imported Employee Records Verified

- Make sure you rearrange the fields accordingly by using Personalise List Columns if required.
- Click on OK.

Result of Initial Data Import:

All the records are successfully imported into the ServiceNow system with Import Set and Transform Map.

Phase 6: Enable Coalesce to Avoid Duplicate Records

- In real projects, the same employee data may be imported multiple times.
- To prevent duplicate employee records, we use Coalesce.
- The Coalesce field will update existing records with new values if re-imported, ensuring duplicate values are not inserted.

Steps:

- **1.** Open Transform Map under System Import Sets / System LDAP in All menus.
- **2.** Open the created Transform map: Sample Spreadsheet Import.
- **3.** Under Related list, select the Field Maps section.
- **4.** Find the u_employee_id field map and set Coalesce = true.
- **5.** Save the form.

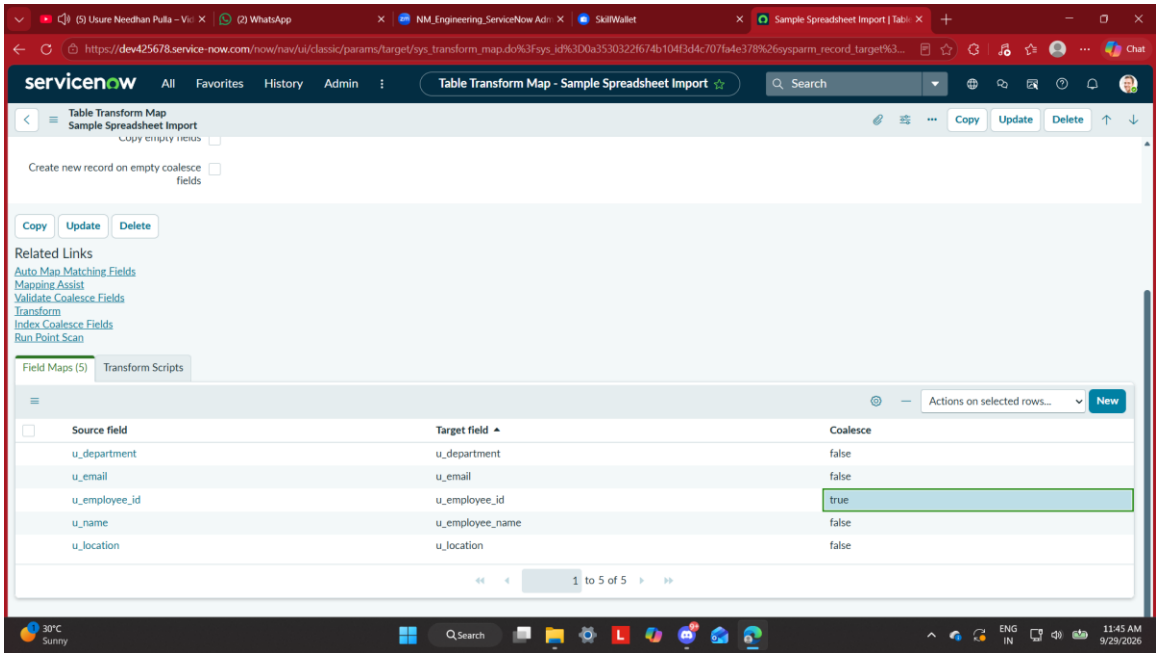


Figure: Table Transform Map - Coalesce Set to True for u_employee_id Field

Phase 7: Inserting New Data In Excel Format

Steps:

- Navigate to All menus.
- Search for Load Data under System Import Sets.
- Fill details:
  - Import set table: Select Existing table {Employee Import}.
  - Source of Import: Select File and choose updated file.
  - Sheet number: 1, Header row: 1.
- Click Submit.

Data Update & Duplicate Verification Scenario:

- In the updated dataset, 4 rows were submitted.
- 2 rows contained existing Employee IDs with updated details (e.g. Email changed from test10@gmail.com to test18@gmail.com, and Employee Name updated).
- 2 rows contained completely new Employee IDs.
- After executing the Transform Map, the transformation progress completed with Success.

Re-running Duplicate Test (Same Spreadsheet Re-Import):

- When the exact same spreadsheet with 4 records was re-imported a second time, Coalesce prevented redundant inserts or updates.
- Checking Transform Histories shows: Total: 4, Inserts: 0, Updates: 0, Ignored: 4.

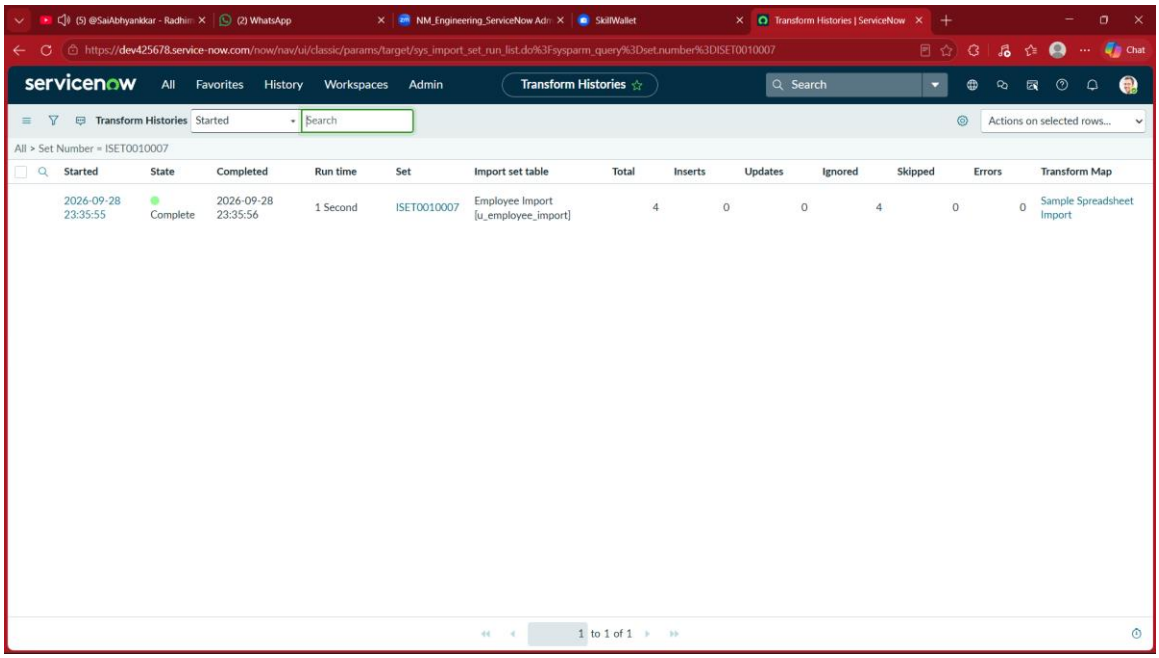


Figure: Transform Histories View - 4 Rows Uploaded, 0 Inserts, 0 Updates, 4 Ignored (Coalesce Working)

- Here we can clearly see that all 4 duplicate rows were correctly Ignored by ServiceNow.

Phase 8: Create Reports on Employee Test Table

Navigation Steps:

- Open All menus, Search for Reports.
- Under Usage and Governance select Reports.
- Click the New button to create a new report.
- Create 3 Reports as follows:

Report 1: Employees by Department

- Report Name: Employees by Department
- Source Type: Table
- Table: Employee Test [u_employee_test]
- Type: Pie Chart
- Configure: Group by = Department, Aggregation = Count
- Style: Default chart colors
- Click Run to execute and Save to store the report.

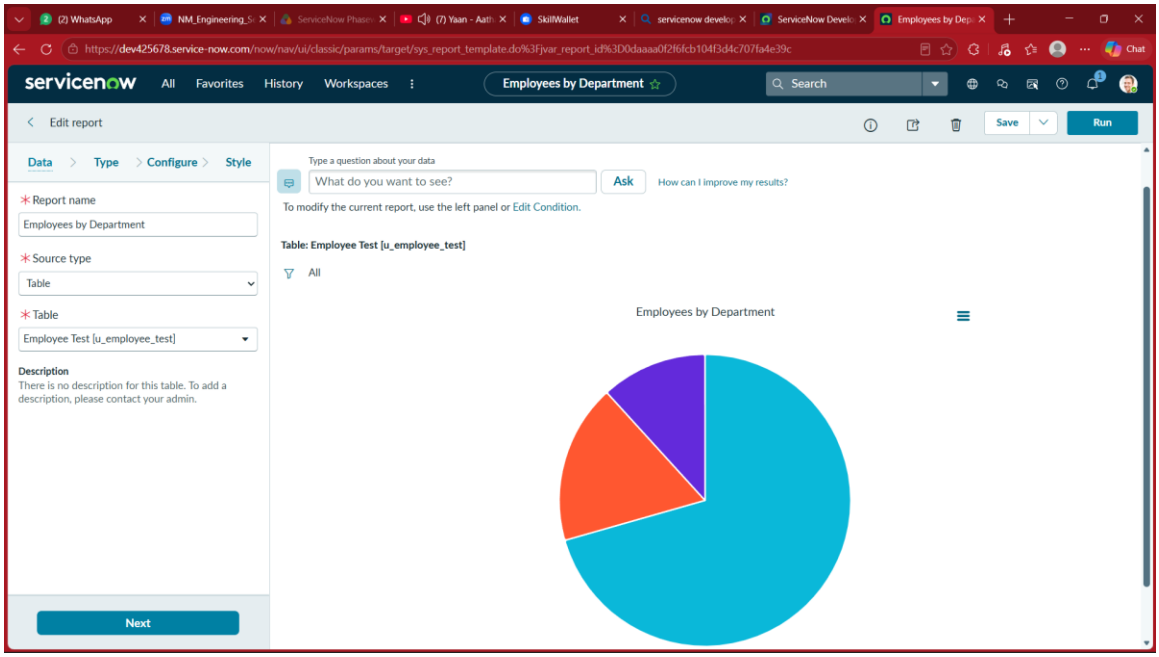


Figure: Report 1: Employees by Department (Pie Chart Configuration & Preview)

Report 2: Employees by Location

- Report Name: Employees by Location
- Source Type: Table
- Table: Employee Test [u_employee_test]
- Type: Bar Chart
- Configure: Group by = Location, Aggregation = Count
- Click Run to execute and Save to store the report.

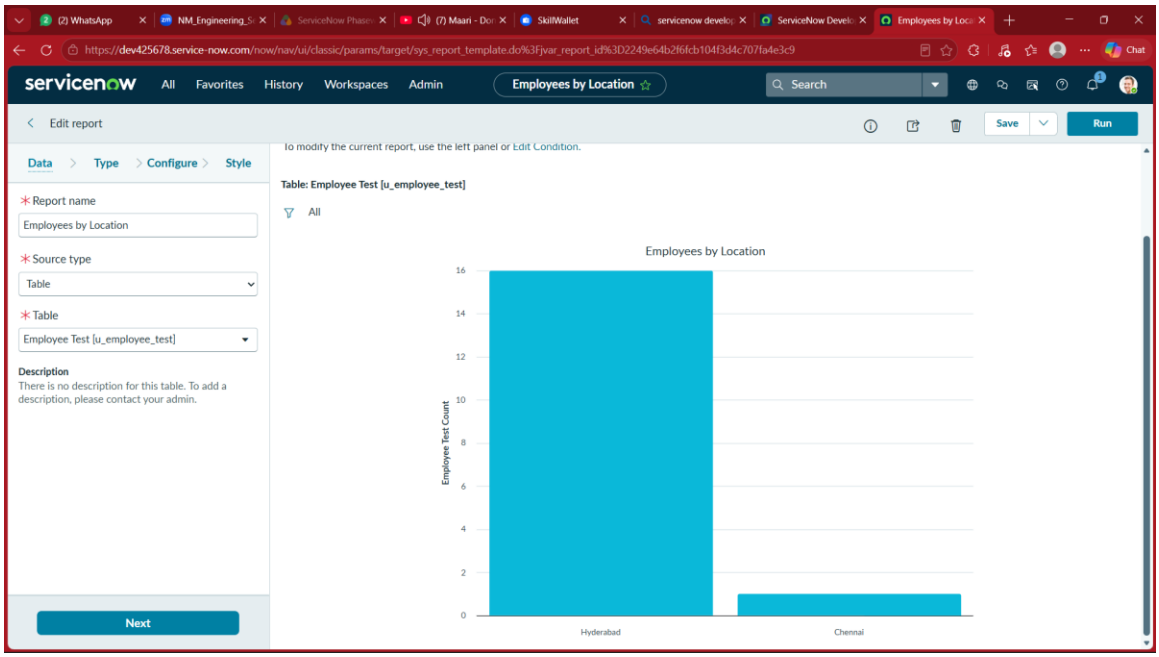


Figure: Report 2: Employees by Location (Bar Chart Configuration & Preview)

Report 3: Employee List Report

- Report Name: Employee List Report
- Source Type: Table
- Table: Employee Test [u_employee_test]
- Type: List
- Configure Select Columns: Employee ID, Employee Name, Email, Department, Location
- Click Run to execute and Save to store the report.

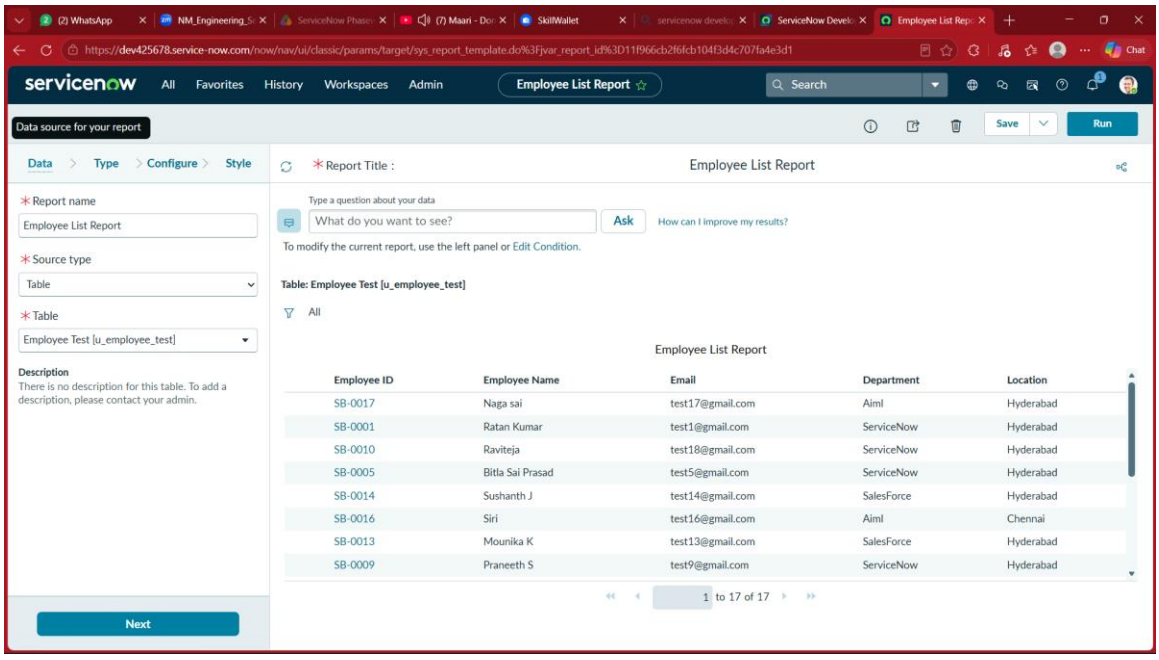


Figure: Report 3: Employee List Report (List View Configuration & Preview)

Phase 9: Add Reports to Dashboard

Create Dashboard

Navigation:

- Open All menus → Search for Dashboards.
- Under Self Service → Click Dashboards.
- Click Create Dashboard or navigate directly using backend form PA_DASHBOARDS.FORM.
- (Note: Backend form access requires ACL pa_dashboards with Admin Overrides set to True).
- Dashboard Name: Employee Analytics Dashboards
- Save the Form.

Add Reports to Dashboard:

- Open each created report (Employees by Department, Employees by Location, Employee List Report).
- Click the Share button and select Add to Dashboards.
- Select Employee Analytics Dashboards and click Add.
- Alternatively, click the '+' icon on the dashboard header to add Report widgets directly.

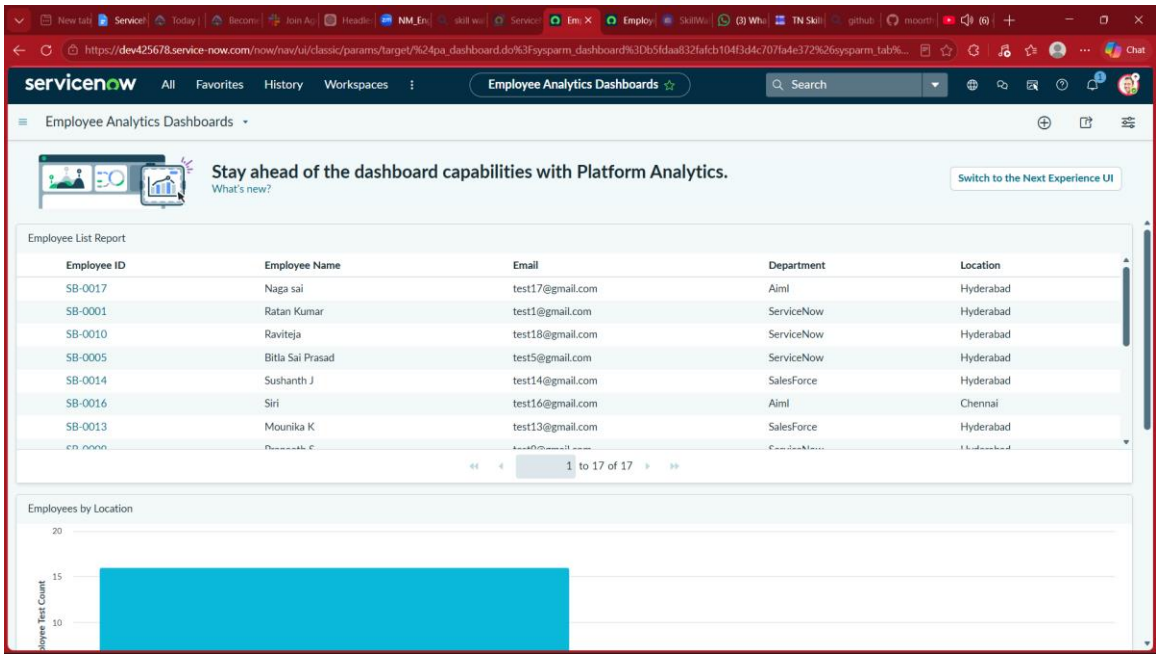


Figure: Employee Analytics Dashboards - View showing Employee List Report & Location Bar Chart

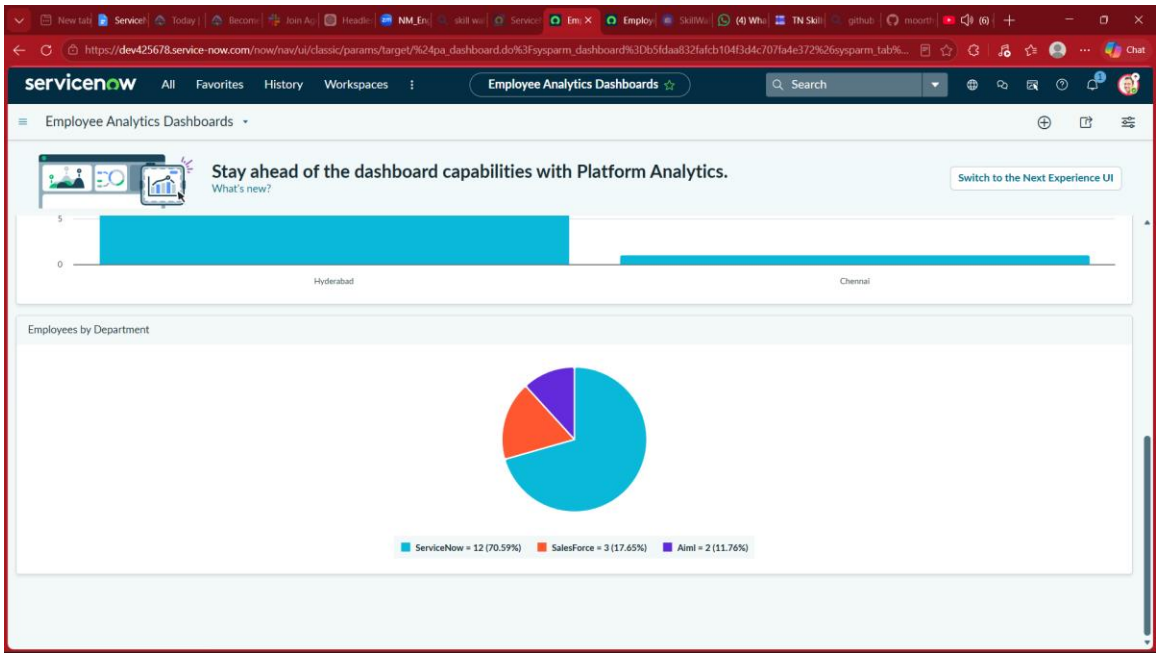


Figure: Employee Analytics Dashboards - View showing Location Bar Chart & Department Pie Chart Widgets

- All three reports are successfully added to the dashboard, providing real-time visibility across Department, Location, and Employee details.
- User, Group, and Role permissions can also be configured via the Dashboard Share panel.

Conclusion

- **1.** This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency, and data integrity throughout the employee onboarding and update process.
- **2.** A key enhancement in this project was the use of Coalesce in the Transform Map. By enabling Coalesce on the Employee ID field, the system intelligently identifies existing employee records during every import. This prevents the creation of duplicate records and ensures that repeated imports update existing data instead of inserting redundant entries. As a result, the employee database remains clean, reliable, and consistent, which is critical in real-world enterprise environments where data may be imported multiple times from external systems.
- **3.** In addition to data integrity, the project was extended with Reports and Dashboards to provide meaningful insights and real-time visibility into employee data and import activities. Custom reports were created to track:
  - Total employee records.
  - Newly added employees.
  - Updated employee records.
  - Department-wise employee distribution.
  - Import success and error counts.

- **4.** These reports were then consolidated into a Dashboard, allowing administrators and HR teams to monitor employee data trends, import performance, and data quality from a single centralized view. Dashboards improved decision-making by presenting information visually through charts and graphs, reducing the need for manual analysis.